ART 106 - FRI

CONTINUOUS

Recall: $z = \dfrac{x - \mu}{\sigma}$

DISCRETE

Problem 1:
You are organizing a conference for next week and believe attendance will depend on the weather. You consider the following probabilities are appropriate:

Weather	**Probability = f(x)**	**Attendance = y**
Terrible weather	0.2	500
Mediocre weather	0.6	1,000
Great weather	0.2	2,000

(a) Let X denote the attendance. Why is X a random variable?
(b) What is the expected attendance?
(c) Suppose each ticket costs \$5 and that the total cost of giving the concert is a fixed \$2,000. Let Y = profit = total sales revenue – total cost = 5,000 – 2,000. What is the expected profit?
(d) If the variance of attendance is $\sigma_X^2 = 240{,}000$, find the variance of profit Y.

(a)

$$E(y) = \sum_{i=1}^{3} y_i P(y_i)$$

(b)

$$= 500(0.2) + 1000(0.6) + 2000(0.2)$$

$$= 1{,}100$$

(c) 3500

(d) 6,000,000

X = TOTAL SALES

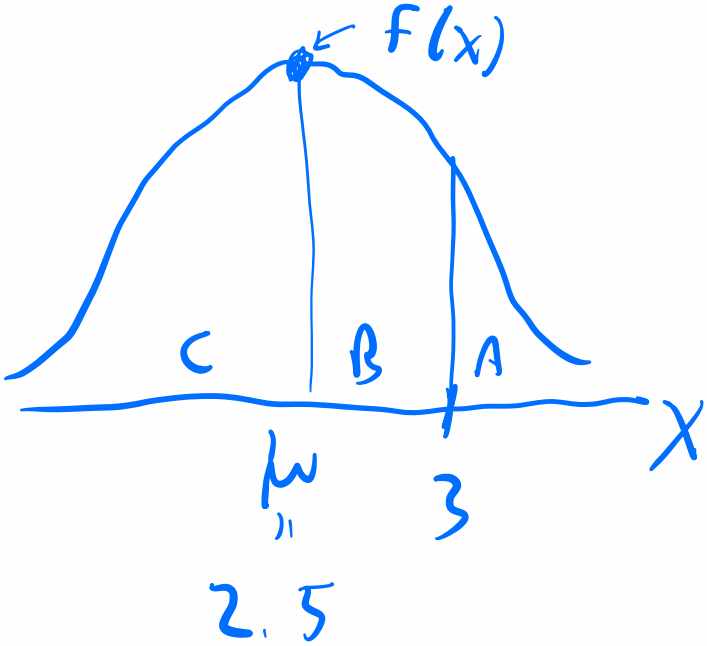
$\mu = 2.5$

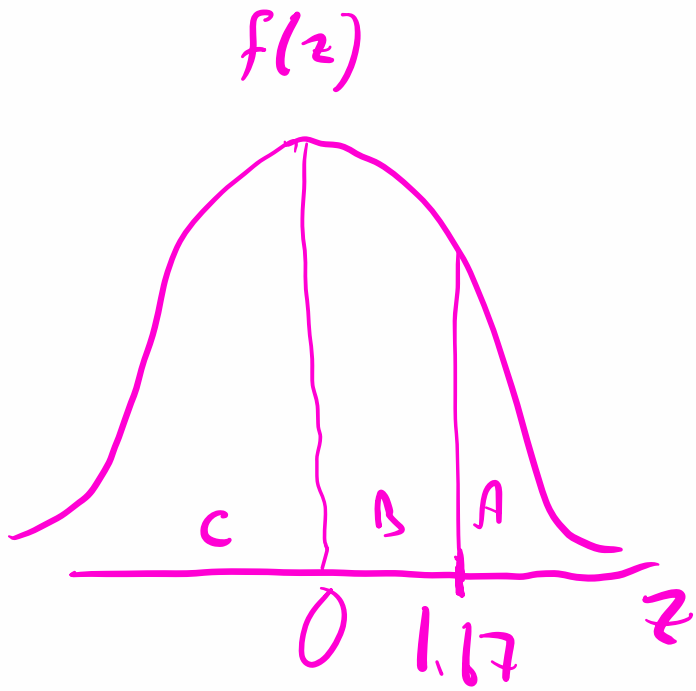
$\sigma = 0.3$

Problem 2:
A firm's marketing manager believes that total sales X can be modelled using normal distribution with mean μ = \$2.5 million and standard deviation σ = \$300,000. What is the probability that sales will exceed \$3 million? Draw a graph

$$X \sim N(2.5, (0.3)^2)$$

$$\boxed{P(X > 3)?}$$





Need B

$$Z = \frac{X - \mu}{\sigma} =$$

$$Z = \frac{3 - 2.5}{0.3} = 1.67$$

A or B = 0.5000

B = 0.4535

A = 0.0465

$P(X > 3)$ 0.0465

$P(X > 3) = 4.15\%$

$Z \sim N(0, 1)$

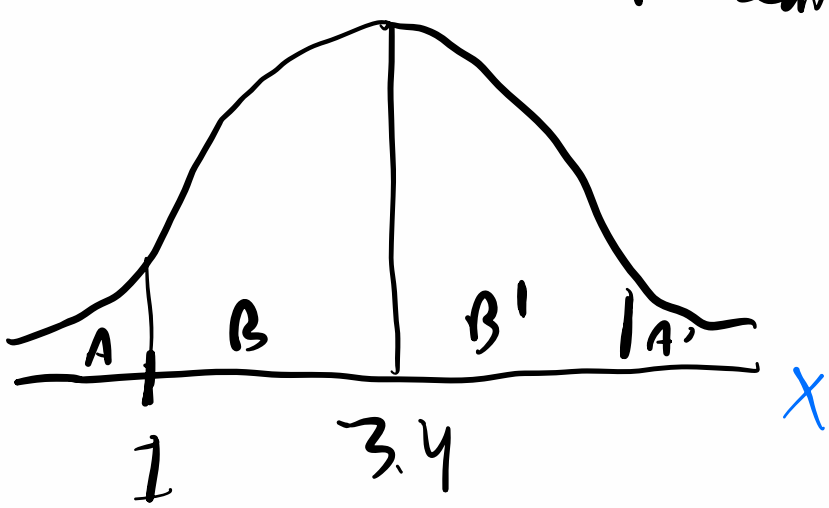
Problem 3:
The length of life (in years) of a personal computer is approximately normally distributed with a mean of 3.4 years and a variance of 1.6.

(a) What fraction of computers will fail in the first year?
(b) What fraction of computers will last at least 2 years?
(c) What fraction of computers will last more than 2.5 years but less than 4 years?
(d) If the manufacturer adopts a warranty policy in which only 5% of the computers have to be replaced, what will be the length of the warranty period?

$X \sim N(3.4, 1.6)$ $\quad \sigma^2 = 1.6$

$\sigma = 1.265$

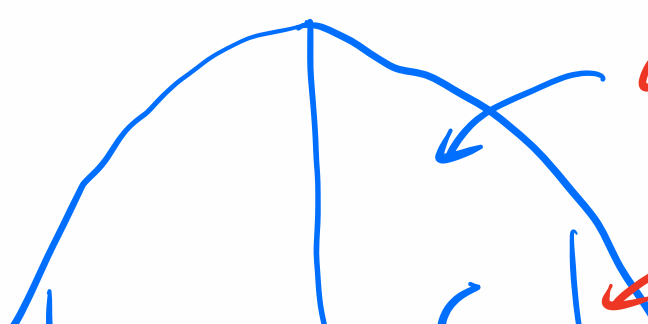
X = years



$\boxed{P(X < 1) = ?}$

we need area A

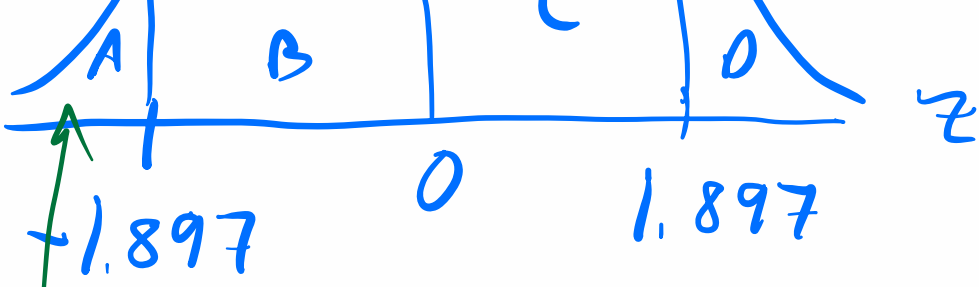
$$Z = \frac{X - \mu}{\sigma} = \frac{1 - 3.4}{\sqrt{1.6}} = -1.897$$



0.4713

AREA D

$0.5 - 0.4713 = 0.0287$



$\boxed{\text{Area A: } 2.87\%}$

(b) $P(x \geq 2) = P(x > 2)$

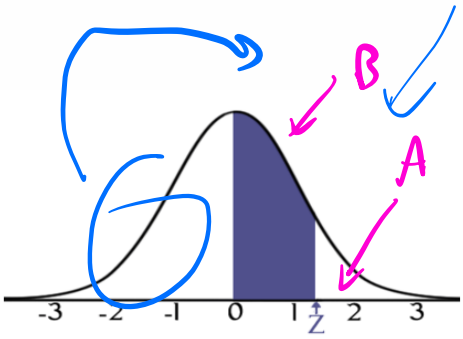
$0.8658 = 86.58\%$

(c) $P(2.5 \leq x \leq 4)$ $\mu = 3.4$

$P = 0.444 = 44.4\%$

(d) IGNORE $x_0 = 1.319$ years

or 15.828 months



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and *z* standard deviations above the mean. For example, for *z* = 1.25 the area under the curve between the mean (0) and *z* is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

(*) BASIC IDEAS OF ~~LINEAR~~

LINEAR REGRESSION

(1) IF 2 variables

$$y_i = \beta_1 + \beta_2 x_i + e_i$$